

Atomic Structure/Periodicity

- In which are the first ionization energies listed in increasing order?
 (A) $N < O < P < S$ (B) $P < S < N < O$
 (C) $P < N < S < O$ (D) $S < P < O < N$
- The Balmer series consists of emission lines that correspond to changes in electron configuration of the hydrogen atom from some initial value of the principal quantum number n to a final value of $n = 2$. The longest-wavelength member of the Balmer series has $\lambda = 656 \text{ nm}$. Which is not a member of the Balmer series?
 (A) 410 nm (B) 434 nm
 (C) 486 nm (D) 545 nm
- The melting points of the group 6 elements increase in the order $\text{Cr} (2180^\circ\text{C}) < \text{Mo} (2896^\circ\text{C}) < \text{W} (3695^\circ\text{C})$. Which is the best explanation for this trend?
 (A) The degree of covalency increases down the group.
 (B) The partial positive charge on the metal atoms in the lattice increases down the group.
 (C) The valence orbitals become increasingly contracted down the group due to relativistic effects.
 (D) The packing density of the metals increases down the group as the lattice changes from simple cubic to body-centered cubic to face-centered cubic.
- Which is the best representation of the lowest-energy excited state of a carbon atom?
 (A) $\begin{array}{ccccc} \uparrow\downarrow & \uparrow & \uparrow & \uparrow & \uparrow \\ | & | & | & | & | \\ 1s & 2s & 2p & & \end{array}$
 (B) $\begin{array}{ccccc} \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow & \uparrow \\ | & | & | & | & | \\ 1s & 2s & 2p & & \end{array}$
 (C) $\begin{array}{ccccc} \uparrow\downarrow & \uparrow\downarrow & \uparrow & & \uparrow \\ | & | & | & & | \\ 1s & 2s & 2p & & \end{array}$
 (D) $\begin{array}{ccccc} \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow & \uparrow \\ | & | & | & | & | \\ 1s & 2s & 2p & & \end{array}$
- Which statements regarding the standard reduction potentials of the group 14 element dioxides XO_2 are correct?
 $\text{XO}_2 + 4 \text{H}^+(aq) + 4 e^- \rightarrow \text{X}(s) + 2 \text{H}_2\text{O}(l) \quad E^\circ(\text{X})$
 I. $E^\circ(\text{C}) < E^\circ(\text{Si})$ II. $E^\circ(\text{Sn}) < E^\circ(\text{Pb})$
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
- By which mode does ^{92}Tc undergo radioactive decay?
 (A) α decay (B) β^- decay
 (C) γ decay (D) Positron emission
- In what group is the atom with the following ionization energies?

IE_1 , kJ mol^{-1}	IE_2 , kJ mol^{-1}	IE_3 , kJ mol^{-1}	IE_4 , kJ mol^{-1}
738	1450	7730	10500

 (A) Group 1 (B) Group 2
 (C) Group 3 (D) Group 13
- In a ground-state gas-phase Sb atom, how many electrons have the quantum numbers $n = 5$, $l = 1$, and $m_l = 0$?
 (A) 1 (B) 2 (C) 3 (D) 6
- Two atomic orbitals have the same number of radial nodes and the same magnetic quantum number m_l . Which other quantum numbers must these orbitals have in common?
 I. The principal quantum number n
 II. The angular momentum quantum number l
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
- The p block elements in the third row of the periodic table (Al–Cl) release more energy when an electron is added to the gas phase atoms than do their lighter congeners (B–F). Which is the best explanation for this phenomenon?
 (A) The heavier elements have more protons in their nuclei and have a greater electrostatic attraction to the electrons.
 (B) The heavier elements have more neutrons in their nuclei and have a greater gravitational attraction to the electrons.
 (C) The $3p$ orbitals are larger and have less electron-electron repulsion.
 (D) The $3p$ orbitals have more nodes and accept electrons at lower energy.
- Which elements are closest in electronegativity?
 (A) C and Al (B) O and Cl
 (C) N and P (D) P and As
- By what mode does ^{212}Po undergo radioactive decay?
 (A) α decay (B) β^- decay
 (C) Positron emission (D) Spontaneous fission

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13. For which gas-phase atoms is addition of an electron endothermic?
- I. Li II. N
- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II
14. Which atom has the lowest first ionization energy?
- (A) Be (B) B (C) Mg (D) Al
15. How many electrons in a ground-state gas-phase atom of arsenic (As) have a quantum number $m_l = 1$?
- (A) 5 (B) 7 (C) 15 (D) 18
16. Which orbital has the same number of radial nodes as a 3s orbital?
- (A) 3p (B) 4s (C) 4f (D) 5d
17. Which lanthanide has a common oxidation number different from +3?
- (A) Ce (B) Gd (C) Er (D) Lu
18. Which radioisotope decays by beta (β^-) emission?
- (A) ^{20}Na (B) ^{52}V (C) ^{54}Co (D) ^{62}Cu
19. One electron in a ground-state gas-phase Ge atom has quantum numbers (n, l, m_l, m_s) of (4, 1, 1, $-\frac{1}{2}$). Which of the following cannot be a valid set of quantum numbers for another electron in this atom?
- (A) (3, 2, 1, $\frac{1}{2}$) (B) (4, 0, 0, $-\frac{1}{2}$)
- (C) (4, 1, 0, $\frac{1}{2}$) (D) (4, 1, -1, $-\frac{1}{2}$)
20. How many nodes does the radial wave function of a 3s orbital have?
- (A) 0 (B) 1 (C) 2 (D) 3
21. Which transition in a gas-phase hydrogen atom would absorb the most energy?
- (A) $n = 1 \rightarrow n = 3$ (B) $n = 2 \rightarrow n = 4$
- (C) $n = 3 \rightarrow n = 1$ (D) $n = 4 \rightarrow n = 2$
22. Which element has a gas-phase electron affinity closest to that of sulfur (S)?
- (A) O (B) P (C) Cl (D) Se
23. The first ionization energy of gallium (Ga, 578.8 kJ mol $^{-1}$) is greater than that of aluminum (Al, 577.5 kJ mol $^{-1}$). Which is the best explanation for this?
- (A) The 3d electrons in gallium imperfectly screen the nuclear charge from the outermost electrons.
- (B) The 4p orbitals in gallium have an additional radial node compared to the 3p orbitals in aluminum and thus have more electron density very close to the nucleus.
- (C) Gallium has a half-filled subshell, which is unusually stable.
- (D) Because electron energy is proportional to $1/n^2$, electrons with higher principal quantum numbers have lower energies and hence are more difficult to ionize.
24. By what mode does the isotope ^{31}S undergo radioactive decay?
- (A) Alpha decay (B) Beta decay
- (C) Gamma decay (D) Positron emission
25. What is the ground-state electron configuration of $\text{Ni}^{2+}(\text{g})$?
- (A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
- (B) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
- (C) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$
- (D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$
26. Which of the following best describes the trend in the amount of energy released on addition of an electron to the gas-phase atoms (electron affinity) of the group 1 and group 2 elements in the same period?
- (A) More energy is released by the group 1 elements because there is less electron-electron repulsion in the ions.
- (B) More energy is released by the group 1 elements because the added electron enters an s orbital rather than a p orbital.
- (C) More energy is released by the group 2 elements because they have more protons in the nucleus.
- (D) More energy is released by the group 2 elements because addition of an electron to a group 1 element disrupts an unusually stable half-filled s subshell.

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27. Two atomic orbitals have the same shape and orientation but have a different number of nodes. Which statement about the quantum numbers of these orbitals is correct?
- (A) They have the same value of l but different values of n .
(B) They have the same value of l but different values of m_l .
(C) They have the same value of m_l but different values of l .
(D) They have the same value of m_s but different values of m_l .
28. A gas-phase hydrogen atom absorbs a photon of visible light and then emits a photon of ultraviolet light. What may be concluded about its initial and its final values of the principal quantum number n ?
- (A) Initially $n = 1$ and finally $n = 2$
(B) Initially $n = 2$ and finally $n = 1$
(C) Initially $n = 2$ and finally $n = 4$
(D) This scenario is impossible because ultraviolet light is more energetic than visible light.
29. How many orbitals are occupied by at least one electron in the ground state of a gas-phase Si atom?
- (A) 3 (B) 5 (C) 7 (D) 8
30. Which transition of a hydrogen atom would involve absorbing a photon with the longest wavelength?
- (A) $n = 1 \rightarrow n = 3$ (B) $n = 2 \rightarrow n = 6$
(C) $n = 3 \rightarrow n = 1$ (D) $n = 6 \rightarrow n = 2$
31. Which gas-phase ion has the greatest number of unpaired electrons in its ground state?
- (A) Cr^{2+} (B) Co^{2+} (C) Ni^{2+} (D) Zn^{2+}
32. The ionization energy of which element is closest to that of fluorine (F)?
- (A) O (B) Ne (C) Cl (D) Ar
33. By what mode does the isotope ^{79}Se undergo radioactive decay?
- (A) Alpha decay (B) Beta decay
(C) Gamma decay (D) Electron capture
34. Despite the fact that atomic radius generally increases with period, the palladium-chlorine distance in PdCl_4^{2-} is essentially identical to the platinum-chlorine distance in PtCl_4^{2-} (2.30 Å). Which is the best explanation of why the two bond lengths are the same?
- (A) The presence of the 14 lanthanides between Pd and Pt makes Pt unexpectedly small.
(B) PdCl_4^{2-} is tetrahedral, while PtCl_4^{2-} is square planar and so has shorter bonds than expected.
(C) The metal-chlorine bonds are highly covalent and so are insensitive to the trend in atomic radius.
(D) The large negative charge on the complex ions makes the chloride-chloride repulsions, not the metal-chloride attractions, the dominant factor in determining the bond distance.
35. A hydrogen atom emits light as it changes from a state with a principal quantum number $n+3$ to a state with principal quantum number n . Which change causes the greatest change in the energy of the emitted photon?
- (A) The initial state is changed to $n+4$.
(B) The initial state is changed to $n+2$.
(C) The final state is changed to $n+1$.
(D) The final state is changed to $n-1$.
36. The first three ionization energies (in kJ mol^{-1}) of four metallic elements are listed below. Which element would form the chloride salt with the lattice energy that is largest in magnitude?
- (A) 419, 3052, 4420 (B) 503, 965, 3600
(C) 520, 7298, 11815 (D) 590, 1145, 4912
37. Which element releases the greatest amount of energy upon addition of an electron to a gas-phase atom?
- (A) N (B) O (C) P (D) S
38. Which species is the strongest oxidant in aqueous solution?
- (A) TcO_4^- (B) RuO_4
(C) ReO_4^- (D) OsO_4
39. Which gas-phase atom contains the greatest number of unpaired electrons in its ground state?
- (A) Be (B) Al (C) Ti (D) Cu

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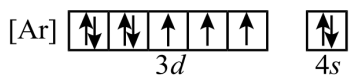
40. A clean sample of magnesium metal is irradiated under vacuum with a 1 W laser. If the laser wavelength is 400 nm, nothing happens, but if the wavelength is 300 nm, then electrons with 45.6 kJ mol^{-1} of energy are ejected from the Mg surface. What happens if a 2 W laser is used instead?
- (A) There is no change at 400 nm, but at 300 nm twice as many electrons are emitted at 45.6 kJ mol^{-1} .
(B) There is no change at 400 nm, but at 300 nm, electrons are emitted at 91.2 kJ mol^{-1} .
(C) There is no change at 400 nm, but at 300 nm, electrons are emitted at 444 kJ mol^{-1} .
(D) At 400 nm, electrons are emitted at 245 kJ mol^{-1} , while at 300 nm, electrons are emitted at 444 kJ mol^{-1} .
41. A hydrogen atom with principal quantum number $n = 3$ can emit light with $\lambda = 656 \text{ nm}$. Which statement about a hydrogen atom with quantum number $n = 4$ is correct?
- (A) It can only emit light with $\lambda < 656 \text{ nm}$.
(B) It can only emit light with $\lambda = 656 \text{ nm}$.
(C) It can only emit light with $\lambda > 656 \text{ nm}$.
(D) It can emit light with wavelengths either less than or greater than 656 nm , but cannot emit light with $\lambda = 656 \text{ nm}$.
42. In which list are the ions ranked in order of increasing ionic radius?
- (A) $\text{K}^+ < \text{Rb}^+ < \text{Ca}^{2+} < \text{Sr}^{2+}$
(B) $\text{K}^+ < \text{Ca}^{2+} < \text{Rb}^+ < \text{Sr}^{2+}$
(C) $\text{Ca}^{2+} < \text{Sr}^{2+} < \text{K}^+ < \text{Rb}^+$
(D) $\text{Ca}^{2+} < \text{K}^+ < \text{Sr}^{2+} < \text{Rb}^+$
43. Which element is the most electronegative?
- (A) N (B) O (C) S (D) Cl
44. How many orbitals are filled or partially filled in a gas-phase Ti atom in its ground state?
- (A) 11 (B) 12 (C) 13 (D) 14
45. ^{41}Ca undergoes radioactive decay by electron capture. Which statements about its decay product are correct?
- I. The decay product is an isotope of Sc.
II. The decay product has a slightly larger mass than ^{41}Ca .
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
46. Aluminum has a smaller first ionization energy than magnesium. Which is the best explanation for this observation?
- (A) Al has an odd number of electrons, while Mg has an even number of electrons.
(B) Al has more valence electrons than Mg.
(C) The highest-energy electron in Al has less electron density near the nucleus than the highest-energy electron in Mg.
(D) The highest-energy electron in Al is on average farther from the nucleus than the highest-energy electron in Mg.
47. In which gas-phase atom are the $2s$ electrons closest on average to the nucleus?
- (A) N (B) O (C) Al (D) Si
48. Which electron configuration corresponds to a gas-phase oxygen atom in an excited state?
- (A) $1s^2 1p^2 2s^2 2p^2$ (B) $1s^2 2p^6$
(C) $1s^2 2s^2 2p^3$ (D) $1s^2 2s^2 2p^4$
49. Alkali metals (Group 1) differ from alkaline earth metals (Group 2) of the same period in which way?
- (A) Alkali metals have larger ionic radii.
(B) Alkali metals have higher melting points.
(C) Alkali metals have greater first ionization energies.
(D) Alkali metals have greater densities.
50. Atoms of which element release the most energy when an electron is added to them in the gas phase?
- (A) C (B) O (C) Si (D) S
51. How many unpaired electrons does a ground-state gas-phase Cr^{2+} ion have?
- (A) 0 (B) 2 (C) 4 (D) 6
52. Which nuclide undergoes beta decay?
- (A) ^{11}C (B) ^{13}N (C) ^{19}O (D) ^{22}Na
53. In a ground-state P atom in the gas phase, how many electrons have quantum numbers $n = 3$, $l = 1$, $m_l = -1$?
- (A) 0 (B) 1 (C) 2 (D) 3

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54. Rank the elements Si, P, Ge, and As in increasing order of their first ionization energies.

(A) $\text{Si} < \text{P} < \text{Ge} < \text{As}$ (B) $\text{As} < \text{Ge} < \text{P} < \text{Si}$
 (C) $\text{Ge} < \text{Si} < \text{As} < \text{P}$ (D) $\text{Ge} < \text{As} < \text{Si} < \text{P}$

55. Which gas-phase atom or ion has the following ground state?



(A) Mn (B) Co (C) Fe^+ (D) Cu^{2+}

56. Which change in principal quantum number for an electron in a hydrogen atom would correspond to emission of the longest-wavelength photon?

(A) $n = 4 \rightarrow n = 1$ (B) $n = 5 \rightarrow n = 2$
 (C) $n = 1 \rightarrow n = 5$ (D) $n = 2 \rightarrow n = 4$

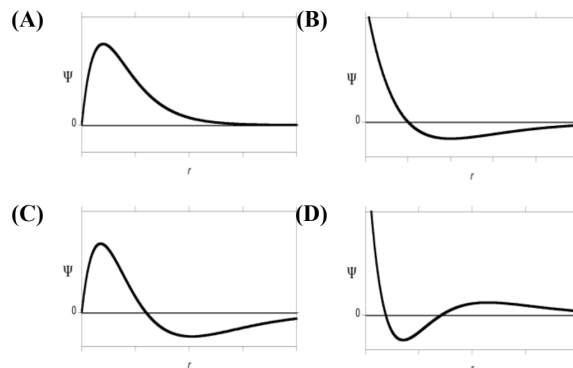
57. Which pair of elements has the most similar electronegativities?

(A) B and C (B) B and Al
 (C) B and Si (D) Al and C

58. The permanganate ion, MnO_4^- , is purple while the pertechnetate ion, TcO_4^- , is colorless. This difference is most closely related to which difference between manganese and technetium?

(A) Manganese(VII) is a stronger oxidant than technetium(VII).
 (B) Manganese consists of a stable isotope while all isotopes of technetium are radioactive.
 (C) The Mn–O bond in permanganate is shorter than the Tc–O bond in pertechnetate.
 (D) Elemental manganese is less dense than elemental technetium.

59. Which graph best describes the radial wavefunction of a $2p$ orbital?



60. In a hydrogen atom, which transition requires the greatest input of energy?

(A) $n = 7 \rightarrow n = 3$ (B) $n = 2 \rightarrow n = 1$
 (C) $n = 3 \rightarrow n = 7$ (D) $n = 1 \rightarrow n = 2$

61. How many unpaired electrons does a ground-state gas-phase Mn^{2+} ion have?

(A) 1 (B) 3 (C) 5 (D) 7

62. Which elements are most similar in atomic size?

(A) H ($Z = 1$) and Li ($Z = 3$)
 (B) C ($Z = 6$) and Si ($Z = 14$)
 (C) Mn ($Z = 25$) and Tc ($Z = 43$)
 (D) Zr ($Z = 40$) and Hf ($Z = 72$)

63. Which halogen has the greatest first ionization energy?

(A) F (B) Cl (C) Br (D) I

64. The isotope ^{69}Zn undergoes what mode of radioactive decay?

(A) Alpha emission (B) Beta emission
 (C) Gamma emission (D) Positron emission

65. Which of the following sets of quantum numbers n , l , m_l correspond to a valence electron in a neutral atom of arsenic (As)?

(A) 3, 0, 0, +1/2 (B) 3, 2, 1, -1/2
 (C) 4, 0, 0, +1/2 (D) 4, 2, 1, -1/2

66. Which gas-phase atom has the largest radius?

(A) Na (B) K (C) Mg (D) Ca

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67. Which of the following does NOT represent the arrangement of electrons in the p subshell in the ground state of any gas-phase atom?
- (A) $\uparrow\downarrow \quad \uparrow \quad \underline{\hspace{0.5cm}}$ (B) $\uparrow \quad \uparrow \quad \uparrow$
 (C) $\uparrow\downarrow \quad \downarrow \quad \downarrow$ (D) $\underline{\hspace{0.5cm}} \quad \underline{\hspace{0.5cm}} \quad \underline{\hspace{0.5cm}}$
68. Which element's electronegativity is closest to that of S?
- (A) O (B) P (C) Cl (D) Se
69. Atomic nitrogen has a higher ionization energy than atomic oxygen. This is best explained by
- (A) the lower electron-electron repulsion in nitrogen.
 (B) the greater effective nuclear charge of nitrogen.
 (C) the fact that the electron ionized in N is from the 2s subshell, while that ionized from O is from the 2p subshell.
 (D) the fact that N has an odd number of electrons while O has an even number.
70. The energy required to break one mole of hydrogen-hydrogen bonds in H_2 is 436 kJ. What is the longest wavelength of light with sufficient energy to break a single hydrogen-hydrogen bond?
- (A) 122 nm (B) 132 nm
 (C) 274 nm (D) 656 nm
71. How many angular nodes does a d orbital possess?
- (A) 1 (B) 2 (C) 3 (D) 4
72. For the transition metal with the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$, how many unpaired electrons are present in its +2 ion in the ground state?
- (A) 0 (B) 2 (C) 4 (D) 6
73. What is the velocity of an electron ($m = 9.11 \times 10^{-28}$ g) that exhibits a de Broglie wavelength of 10.0 nm? [1 J = 1 kg•m²•s⁻²]
- (A) 72.7 m•s⁻¹ (B) 270 m•s⁻¹
 (C) 7.27×10^4 m•s⁻¹ (D) 7.27×10^6 m•s⁻¹
74. Which atom has the highest first ionization energy?
- (A) Al (B) Si (C) P (D) S
75. What are the two most common oxidation states for antimony?
- (A) -1 and -3 (B) -1 and +2
 (C) +3 and -3 (D) +3 and +5
76. For which pair of species are the radii most similar?
- (A) Li and Na (B) Na and Mg
 (C) Mn and Fe (D) Fe^{2+} and Fe^{3+}
77. Which is an acceptable set of quantum numbers for an electron?
- (A) 1, 1, 0, 1/2 (B) 2, 1, 0, 0
 (C) 2, 1, -1, -1/2 (D) 3, 2, -3, 1/2
78. Which orbital has the most radial nodes?
- (A) 4s (B) 4p (C) 4d (D) 4f
79. Ionization energies vary from left to right across the periodic table. Factors that contribute to this variation include which of the following?
- I. changes in the nuclear charge
 II. differences in shielding by valence electrons
 III. differences in shielding by core electrons
- (A) I. only (B) III. only
 (C) I. and II. only (D) I., II. and III.
80. Which gaseous ion has exactly three unpaired electrons?
- (A) Fe^{3+} (B) Ni^{2+} (C) Ti^{4+} (D) V^{2+}
81. Which element has the highest second ionization energy?
- (A) Na (B) Mg (C) S (D) F
82. The ions below are listed in order of increasing radii except
- (A) $Al^{3+} < Mg^{2+} < Na^+$ (B) $K^+ < Cl^- < Se^{2-}$
 (C) $Cl^- < Br^- < I^-$ (D) $O^{2-} < F^- < Na^+$
83. How many radial nodes does a 3d orbital possess?
- (A) 0 (B) 1 (C) 2 (D) 3
84. The successive ionization energies (in kJ/mol) for an element are shown below.
- | E_1 | E_2 | E_3 | E_4 | E_5 |
|-------|-------|-------|-------|-------|
| 577 | 1820 | 2740 | 11600 | 14800 |
- What is the electron configuration of this element?
- (A) $1s^2 2s^2 2p^6 3s^1$ (B) $1s^2 2s^2 2p^6 3s^2 3p^1$
 (C) $1s^2 2s^2 2p^6 3s^2 3p^3$ (D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$

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85. Albert Einstein's explanation of the photoelectric effect confirmed which of the following concepts?
- (A) Electrons can absorb energy and change levels in atoms.
(B) Light energy can be converted into the mass of electrons.
(C) Electrons have both particle and wave properties.
(D) Light has both particle and wave properties.
86. Which gas phase ion in its ground state has the greatest number of unpaired electrons?
- (A) Cr^{3+} (B) Mn^{3+} (C) Fe^{3+} (D) Co^{3+}
87. For the element with the electron configuration $1s^2 2s^2 2p^6 3s^2$, one of the 3s electrons will be shielded from the nuclear charge most effectively by a
- (A) 1s electron (B) 2s electron
(C) 2p electron (D) 3s electron
88. In which list are atoms of the elements Be, B, Mg and Al arranged from smallest to largest atomic radius?
- (A) $\text{Be} < \text{B} < \text{Mg} < \text{Al}$ (B) $\text{Mg} < \text{Be} < \text{Al} < \text{B}$
(C) $\text{B} < \text{Be} < \text{Al} < \text{Mg}$ (D) $\text{Al} < \text{Mg} < \text{B} < \text{Be}$
89. Which gas-phase atom in its ground state has exactly three unpaired electrons?
- (A) Sc (B) Fe (C) Co (D) Se
90. The first ionization energy of Na is $495.9 \text{ kJ}\cdot\text{mol}^{-1}$. What is the longest wavelength of light that could remove an electron from a Na atom?
- (A) $2.41 \times 10^{-7} \text{ m}$ (B) $2.41 \times 10^{-4} \text{ m}$
(C) 4.14 m (D) $4.14 \times 10^3 \text{ m}$
91. Which set of quantum numbers could represent an electron in a 5f orbital?
- (A) $l = 4, m_l = 2$ (B) $l = 2, m_l = -3$
(C) $l = 3, m_l = 4$ (D) $l = 3, m_l = 0$
92. Which orbital possesses one angular node and one radial node?
- (A) 2s (B) 2p (C) 3p (D) 3d
93. The radius of which ion is closest to that of the Li^+ ion?
- (A) Na^+ (B) Be^{2+} (C) Mg^{2+} (D) Al^{3+}
94. Of the atoms listed, which has the largest third ionization energy?
- (A) Ca (B) Mg (C) Al (D) Si
95. Which set of quantum numbers (n, l, m_l, m_s) is possible for the outermost electron in a strontium atom in its ground state?
- (A) 5, 0, 0, $-1/2$ (B) 5, 0, 1, $1/2$
(C) 5, 1, 0, $1/2$ (D) 5, 1, 1, $-1/2$
96. How many orbitals are in an f sublevel ($l = 3$)?
- (A) 3 (B) 5 (C) 7 (D) 14
97. What is the energy of photons with a wavelength of 434 nm?
- (A) $2.76 \times 10^5 \text{ kJ/mol}$ (B) $2.76 \times 10^2 \text{ kJ/mol}$
(C) $2.76 \times 10^{-1} \text{ kJ/mol}$ (D) $2.76 \times 10^{-4} \text{ kJ/mol}$
98. In which choice are the species listed in order of increasing radius?
- (A) $\text{Na}^+, \text{Mg}^{2+}, \text{Al}^{3+}$ (B) $\text{Cl}^-, \text{S}^{2-}, \text{P}^{3-}$
(C) $\text{Ar}, \text{K}^+, \text{Cl}^-$ (D) $\text{Cl}^-, \text{Ar}, \text{K}^+$
99. Which element has the highest melting point?
- (A) Na (B) K (C) Mg (D) Ca
100. Which element forms a compound with the formula H_3XO_4 ?
- (A) As (B) Cl (C) N (D) S
101. What is the frequency of light with a wavelength of 480 nm?
- (A) $1.60 \times 10^{-6} \text{ s}^{-1}$ (B) $6.25 \times 10^5 \text{ s}^{-1}$
(C) $6.25 \times 10^{14} \text{ s}^{-1}$ (D) $1.44 \times 10^{20} \text{ s}^{-1}$
102. Which set of quantum numbers (n, l, m_l) is forbidden?
- (A) 3, 2, 0 (B) 3, 1, -1
(C) 2, 0, 0 (D) 1, 1, 0
103. Which characteristic of an atomic orbital is most closely associated with the magnetic quantum number, m_l ?
- (A) size (B) shape
(C) occupancy (D) orientation

Atomic Structure/Periodicity

104. Which element exhibits the successive ionization energies given in the table?

Ionization Energy, $\text{kJ}\cdot\text{mol}^{-1}$	
1 st	738
2 nd	1451
3 rd	7733
4 th	10540
5 th	13628

- (A) Na (B) Mg (C) Al (D) Si
105. The energies of the 3s, 3p and 3d orbitals in a multi-electron atom increase in that order. To which factor(s) can this order be attributed?
- I. the relative penetrations of these orbitals near the nucleus

II. the relative average distance of the electron from the nucleus
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
106. Which element has the highest first ionization energy?
- (A) Al (B) Si (C) P (D) S
107. Which set of quantum numbers corresponds to an electron in a 4d orbital?
- (A) $n = 4, \ell = 1, m_\ell = -1, m_s = 1/2$
(B) $n = 4, \ell = 2, m_\ell = -2, m_s = -1/2$
(C) $n = 4, \ell = 3, m_\ell = 3, m_s = 1/2$
(D) $n = 4, \ell = 3, m_\ell = -1, m_s = -1/2$
108. What is the energy of a photon from a laser that emits light at 632.8 nm?
- (A) $3.14 \times 10^{-19} \text{ J}$ (B) $1.26 \times 10^{-31} \text{ J}$
(C) $2.52 \times 10^{-33} \text{ J}$ (D) $4.19 \times 10^{-40} \text{ J}$
109. How many unpaired electrons are in a gaseous Co^{2+} ion in its ground state?
- (A) 1 (B) 3 (C) 5 (D) 7
110. Which ion is **not** isoelectronic with Ar?
- (A) S^{2-} (B) K^+ (C) Sc^{2+} (D) Ti^{4+}
111. Which process releases the most energy?
- (A) $\text{Mg}^{2+}(\text{g}) + \text{e}^- \rightarrow \text{Mg}^+(\text{g})$
(B) $\text{Mg}^+(\text{g}) + \text{e}^- \rightarrow \text{Mg}(\text{g})$
(C) $\text{Na}^{2+}(\text{g}) + \text{e}^- \rightarrow \text{Na}^+(\text{g})$
(D) $\text{Na}^+(\text{g}) + \text{e}^- \rightarrow \text{Na}(\text{g})$

112. In which list are the ions arranged in order of increasing size?

- (A) $\text{F}^- < \text{S}^{2-} < \text{Al}^{3+} < \text{Mg}^{2+}$
(B) $\text{F}^- < \text{S}^{2-} < \text{Mg}^{2+} < \text{Al}^{3+}$
(C) $\text{Mg}^{2+} < \text{F}^- < \text{Al}^{3+} < \text{S}^{2-}$
(D) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{F}^- < \text{S}^{2-}$

113. Which is the symbol for an element whose ground state atoms have the same total numbers of s electrons and p electrons?

- (A) ${}_5\text{B}$ (B) ${}_6\text{C}$ (C) ${}_{12}\text{Mg}$ (D) ${}_{18}\text{Ar}$

114. Which set of quantum numbers is NOT allowed?

n	ℓ	m_ℓ	m_s
(A) 1	0	0	—
(B) 2	2	1	—
(C) 3	1	1	—
(D) 4	3	-3	—

115. Which change(s) in electron structure occur when a gas phase Mn atom is converted to a Mn^{2+} ion in the gas phase?

- I. The number of occupied energy levels decreases.

II. The number of half-filled orbitals decreases.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

116. Which list gives the symbols of the elements in the order of increasing first ionization energy?

- (A) F, Ne, Na (B) Al, Mg, Na
(C) Sr, Ca, Mg (D) Cl, Br, I

117. How many unpaired electrons are in a gas phase Co^{2+} ion in its ground state?

- (A) 2 (B) 3 (C) 4 (D) 5

118. The energy required to ionize a potassium ion is $419 \text{ kJ}\cdot\text{mol}^{-1}$. What is the longest wavelength of light that can cause this ionization?

- (A) 285 nm (B) 216 nm
(C) 200 nm (D) 107 nm

119. When $\ell = 3$, what are the possible values for the quantum number m_ℓ ?

- (A) 2, 1, 0 (B) 3, 2, 1, 0
(C) 2, 1, 0, -1, -2 (D) 3, 2, 1, 0, -1, -2, -3

Atomic Structure/Periodicity

120. The first ionization energy of cesium is 6.24×10^{-19} J/atom. What is the minimum frequency of light that is required to ionize a cesium atom?
- (A) $1.06 \times 10^{15} \text{ s}^{-1}$ (B) $4.13 \times 10^{14} \text{ s}^{-1}$
 (C) $9.42 \times 10^{14} \text{ s}^{-1}$ (D) $1.60 \times 10^{18} \text{ s}^{-1}$
121. When the isoelectronic ions, Cl^- , S^{2-} and K^+ are arranged in order of increasing size, which order is correct?
- (A) K^+ , Cl^- , S^{2-} (B) K^+ , S^{2-} , Cl^-
 (C) S^{2-} , Cl^- , K^+ (D) Cl^- , S^{2-} , K^+
122. Which is most similar for the elements in a group in the periodic table?
- (A) physical state
 (B) melting point
 (C) first ionization energy
 (D) ground state electron configuration
123. How many unpaired electrons are present in a gaseous Co^{3+} ion in its ground state?
- (A) 1 (B) 3 (C) 4 (D) 5
124. Which nucleus is not radioactive?
- (A) K-38 (B) K-39 (C) K-42 (D) K-43
125. How many orbitals are in an atomic sublevel with $l = 3$?
- (A) 3 (B) 5 (C) 7 (D) 9
126. A ground state gaseous atom of which element has the greatest number of unpaired electrons?
- (A) As (B) Br (C) Ge (D) Se
127. An atom of which element has the highest second ionization energy?
- (A) Na (B) Mg (C) Al (D) K
128. Which of these properties increase across the period from Na to Cl?
- | | |
|------|-------------------|
| I. | atomic radius |
| II. | density |
| III. | electronegativity |
- (A) I only (B) III only
 (C) I and II only (D) II and III only
129. For the elements in group 14 (C to Pb), which property increases with increasing atomic number?
- (A) melting points
 (B) covalent radius
 (C) magnitude of stable oxidation state
 (D) ability to form chains of atoms with themselves
130. What mode of radioactive decay is most likely for the isotope $^{20}_{11}\text{Na}$?
- (A) alpha (B) beta
 (C) gamma (D) electron capture
131. Which experimental evidence most clearly supports the suggestion that electrons have wave properties?
- (A) diffraction
 (B) emission spectra
 (C) photoelectric effect
 (D) deflection of cathode rays by a magnet
132. Which quantum number determines the number of angular nodes in an atomic orbital?
- (A) n (B) l (C) m_l (D) m_s
133. Which element exhibits the greatest number of oxidation states in its compounds?
- (A) Ca (B) V (C) Cu (D) As
134. Of the elements given, which has the lowest ionization energy?
- (A) N (B) P (C) S (D) Cl
135. How many unpaired electrons are in a gaseous Fe^{2+} ion in its ground state?
- (A) 0 (B) 2 (C) 4 (D) 6
136. Which species is most likely to lose a positron (β^+)?
- (A) $^{12}_7\text{N}$ (B) $^{18}_8\text{O}$ (C) $^{20}_9\text{F}$ (D) $^{20}_{10}\text{Ne}$
137. For which compound does the reaction,
 $\text{MCO}_3(\text{s}) \rightarrow \text{MO}(\text{s}) + \text{CO}_2(\text{g})$
 occur most readily?
- (A) BeCO_3 (B) MgCO_3
 (C) CaCO_3 (D) BaCO_3

Atomic Structure/Periodicity

138. Which region of the electromagnetic spectrum is capable of inducing electron transitions with the greatest energy?
(A) infrared (B) microwave
(C) ultraviolet (D) visible
139. All of the following possess complete d shells EXCEPT
(A) Ag^+ (B) Cu^{2+} (C) Ga^{3+} (D) Zn^{2+}
140. Which orbital fills completely immediately before the 4f?
(A) 6s (B) 5p (C) 5d (D) 4d
141. Which set of quantum numbers (n, l, m_l, m_s) is permissible for an electron in an atom?
(A) 1, 0, 0, $-1/2$ (B) 1, 1, 0, $+1/2$
(C) 2, 1, 2, $+1/2$ (D) 3, 2, -2, 0
142. When the elements Li, Be, and B, are arranged in order of increasing ionization energy, which is the correct order?
(A) Li, B, Be (B) B, Be, Li
(C) Be, Li, B (D) Li, Be, B
143. Which forms the most alkaline solution when added to water?
(A) Al_2O_3 (B) B_2O_3 (C) CO_2 (D) SiO_2
144. Which element commonly exhibits both +1 and +3 oxidation states?
(A) Al ($Z = 13$) (B) Sc ($Z = 21$)
(C) Sn ($Z = 50$) (D) Tl ($Z = 81$)
145. In which list are the elements arranged in order of increasing first ionization energy?
(A) Li, Na, K (B) S, O, F
(C) Na, Mg, Al (D) F, Ne, Na
146. Which quantum number is associated with the shape of an atomic orbital?
(A) n (B) l (C) m_l (D) m_s
147. Consider the ions Li^+ , Na^+ , Be^{2+} , and Mg^{2+} . Which two are closest to one another in size?
(A) Li^+ and Na^+ (B) Be^{2+} and Mg^{2+}
(C) Be^{2+} and Li^+ (D) Li^+ and Mg^{2+}
148. What is the electron configuration for a gas phase +3 ion of iron ($Z = 26$)?
(A) $[\text{Ar}] 3d^5$ (B) $[\text{Ar}] 4s^2 3d^3$
(C) $[\text{Ar}] 4s^1 3d^4$ (D) $[\text{Ar}] 4s^2 3d^6$
149. Magnesium ($Z = 12$) has isotopes that range from Mg-20 to Mg-31. Only Mg-24, Mg-25, and Mg-26 are not radioactive. What mode of radioactive decay would convert Mg-20, Mg-21, Mg-22, and Mg-23 into stable isotopes most quickly?
(A) electron emission (B) alpha particle emission
(C) gamma emission (D) positron emission
150. Which element melts at the highest temperature?
(A) aluminum (B) silicon
(C) phosphorus (D) sulfur
151. Which property of an element is most dependent on the shielding effect?
(A) atomic number
(B) atomic mass
(C) atomic radius
(D) number of stable isotopes
152. How many unpaired electrons are present in a ground state gaseous Ni^{2+} ion?
(A) 0 (B) 2 (C) 4 (D) 6
153. When the elements carbon, nitrogen and oxygen are arranged in order of increasing ionization energies, what is the correct order?
(A) C, N, O (B) O, N, C
(C) N, C, O (D) C, O, N
154. Given this set of quantum numbers for a multi-electron atom: 2, 0, 0, $1/2$ and 2, 0, 0, $-1/2$. What is the next higher allowed set of n and l quantum numbers for this atom in its ground state?
(A) $n = 2, l = 0$ (B) $n = 2, l = 1$
(C) $n = 3, l = 0$ (D) $n = 3, l = 1$
155. Which element will exhibit the photoelectric effect with light of the longest wavelength?
(A) K (B) Rb (C) Mg (D) Ca
156. All these elements have common allotropes *except*
(A) C (B) O (C) Kr (D) S

Atomic Structure/Periodicity

157. How many unpaired electrons are in a gaseous Fe^{2+} ion in the ground state?
(A) 0 (B) 2 (C) 4 (D) 6
158. Which element has the smallest first-ionization energy?
(A) Mg (B) Al (C) Si (D) P
159. Which set of orbitals is listed in the sequential order of filling in a many-electron atom?
(A) $3s, 3p, 3d$ (B) $3d, 4s, 4p$
(C) $3d, 4p, 5s$ (D) $4p, 4d, 5s$
160. Which set is expected to show the smallest difference in first-ionization energy?
(A) He, Ne, Ar (B) B, N, O
(C) Mg, Mg^+ , Mg^{2+} (D) Fe, Co, Ni
161. When the atoms Li, Be, B, and Na are arranged in order of increasing atomic radius, what is the correct order?
(A) B, Be, Li, Na (B) Li, Be, B, Na
(C) Be, Li, B, Na (D) Be, B, Li, Na
162. All of these sets of quantum numbers are permissible *except*
- | | n | l | m_l | m_s |
|-----|-----|-----|-------|-------|
| (A) | 1 | 0 | 0 | +1/2 |
| (B) | 2 | 2 | 0 | -1/2 |
| (C) | 3 | 1 | 1 | -1/2 |
| (D) | 3 | 2 | -1 | +1/2 |
163. Which element can exhibit more than one oxidation state in compounds?
1. Cr 2. Pb 3. Sr
(A) 1 only (B) 1 and 2 only
(C) 2 and 3 only (D) 1, 2 and 3
164. When the isoelectronic species, K^+ , Ca^{2+} , and Cl^- , are arranged in order of increasing radius, what is the correct order?
(A) K^+ , Ca^{2+} , Cl^- (B) K^+ , Cl^- , Ca^{2+}
(C) Cl^- , Ca^{2+} , K^+ (D) Ca^{2+} , K^+ , Cl^-
165. Which Group 2 element has chemical properties least like the other members of the group?
(A) Be (B) Ca (C) Sr (D) Ba
166. In the vapor state which atom has the largest ionization energy?
(A) Na (B) K (C) Mg (D) Ca